The Digital Humanities Coursebook

An Introduction to Digital Methods for Research and Scholarship

Johanna Drucker

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4 Metadata, markup, and data description

4a Metadata and classification

The concept of data might be familiar, but the term *metadata* is likely to be less so. Metadata suggests specialized expertise and a world of professional knowledge—for good reason. But it is also a part of files and documents and is an important aspect of doing research that involves description and classification of digital assets. Think of metadata as the information about a resource or digital asset. When you open the "Get Info" box on any file, it will tell you the date, size, and format of the file. That is metadata. Library records and museum catalogue entries are metadata. Metadata is structured by fields like creator, date, place, medium, format, and other information. For digital resources, two levels of metadata may be required—information about the original source (e.g. a painting or film) and the digital file (who made it, when, in what format, and so on).

Metadata ranges in complexity from very basic (like what is displayed when "Get Info" is clicked for any file) to highly complex (descriptions of where, when, under what circumstances, by whom, for whom, and so on) something was made, acquired, damaged, loaned, or engaged in any process that could be recorded. Every researcher working in digital assets needs familiarity with basics.

General points about metadata

Metadata performs many functions. Most often, as in the example of "Get Info" just given, it is *descriptive*. Metadata can also be *administrative*, helping to organize the data in large sets of records. In addition, it can be *operational*, giving data a role or task, such as the relation of one file to another. The differences among these functions will be discussed later in this section. Metadata has a powerful role in every conceivable discipline that produces information (this is true whether that is in analog or digital form).

Metadata is an essential aspect of any digital scholarship or research. Data have to be described and identified in order to be useful. Without

metadata, information in files would be like books without covers or title pages on shelves without labels. Thinking about metadata and designing metadata fields and schemes is heady stuff. And it is where the intellectual and conceptual modeling of research projects takes place.

Metadata is closely related to classification systems and standards. Related concepts include controlled vocabulary, and relations between taxonomies (systems of names) and ontologies (schemes of knowledge organization). These are all terms with enormous history and are also fraught with ethical issues that are the subject of ongoing and essential debates.

Understanding metadata

To reiterate, metadata is information used to describe data, objects, or records. Metadata can be attached to objects or records, but it also stands in as a surrogate. For instance, one common form of metadata is a library record which has fields to describe books and other objects (newspapers, maps, journals, audio-visual materials, etc.). The standard library record has defined fields in which information about the object is specified. The goal of the record is to help patrons locate materials that suit their interests. The library record is also a thing in itself, as well as being a surrogate or stand-in for the object. When surrogates become part of digital systems, their capacity to be analyzed increases as a result of the many benefits of automated processing. But for this to be true, standardization of the organization as well as the substance and format of the information has to occur.

Descriptive metadata is very common. Systems of naming, identifying, and describing objects or entities (including data) are called metadata schemes. Metadata is integral to all systems and activities of knowledge classification. It is essential for organization and access. Metadata is powerful because it frames information in classification systems and terminology. Imagine data as information in any form sitting on shelves, in drawers, or boxes (objects, numbers, or files). Metadata provides the labels for those entities and the organizational system in which they are stored. Think of a vast warehouse for a history museum that has collections from prehistoric settlements to contemporary consumer objects. What system of naming, organizing, and classifying these materials will describe what is what, where is it, and how will you find it? Decisions about how things are described involve challenges of all kinds, since to a great extent, this depends on the anticipated use of those things—and the values attached to them.

Administrative metadata helps to sort data and records according to their use or type. While a catalog record for a book contains certain kinds of information, another set of records might track loans, conservation and repair history, acquisition, provenance, or other features for specialized purposes (that it is part of a particular donor's collection). Administrative

metadata might be identified by the type or function it serves within the larger record keeping structure. If descriptive metadata helps to provide information about objects or data, administrative metadata might identify items that have privacy issues attached or should only be shown to authorized individuals. Metadata that is relevant to institutional activity is different in type from metadata describing information or objects.

Operational metadata provides information on the functional requirements—what is needed for its processing or display. It may contain instructions specifying the resolution of an image as a thumbnail, a screen image, or an archival file, or the system needed to make an audio file play. This metadata assists the system in managing and delivering assets as well as describing them.

As already noted, metadata can describe an object or its representation. So, if you have a digital file of a glass plate photograph of a temple in Athens, taken in 1902, does the metadata describe the photograph's creator, time and date, the camera and glass plate, or the temple that is depicted? The photograph is already a surrogate. The digital asset made from the photograph is a surrogate of that photograph. All of this needs to be recorded in the metadata.

Metadata standards

Metadata standards are essential in all disciplines. But they are often highly contested and even controversial because they embody value judgments either implicitly or explicitly. Do all works, for instance, have "creators" or "authors"? Stories and songs from Indigenous communities might not. As concepts of metadata change, older standards come up for debate.

One common metadata standard is MARC (Machine Readable Cataloging) designed in the early years (1960s) of automated library systems' standardized descriptions. MARC was originally used in local automated systems but was integrated into online catalogs as these emerged—thus the benefit of its original design as a machine-readable format. BIBFRAME, a new standard, was launched by the Library of Congress in about 2011, as a proposed replacement for MARC. BIBFRAME contains features deliberately designed for networked environments. These are standards that are used globally, not just in the United States.

Many objects and documents need metadata besides books and bibliographical artifacts. For the fields of cultural heritage materials, CCO (Cataloging Cultural Objects); AAT (Art and Architecture Thesaurus); the Getty Vocabularies (for works of art, but also their surrogates—like photographic documentation—and conservation); UDS (Universal Decimal System), an adaptation of the Dewey system used internationally in many libraries); and CERIF (Common European Research Information Format). Every field you can think of (from astronomy to zoology) has metadata standards associated

with its record-keeping practices. Research in many scientific areas is coded into these standards to make the work of individual teams useful across projects. Consider the value of weather data collected and standardized for centuries so it can be used to examine changes in patterns at a global level. Without metadata to identify its source, location, and standards, it will not be useful. Imagine the challenges with gathering weather data when time zones were not linked, measurements were only taken on the ground, and reporting was sparse, intermittent, and distributed. Metadata standards, like data standards, are crucial for aggregation.

Humanities metadata

Many metadata schemes for what is referred to as "resource description" follow the standard known as Dublin Core, originally created in the mid-1990s. This is composed of a minimal set of fields for describing analog or digital resources, but with attention to features like encoding standards relevant to the latter. Learning what is meant by the terms and elements in this standard is a good exercise. [See Exercise #1: Standard metadata.]

Many fields might seem obvious, like "creator," "date," "format," or "language." Others might be less clear, such as "coverage" or "contributor" and "type." Because they are widely used, the Dublin Core metadata standards are a good place to begin a humanities project. Consider which aspects of a specialized research project would not be accommodated in this scheme.

The descriptive fields contain information that stands in for or describes the object. However, more advanced description might be necessary, such as administrative fields to indicate what kind of record this is ("license" or

Table 4.1 Dublin Core fields

Contributor
Coverage
Creator
Date
Description
Format
Identifier
Language
Publisher
Relation
Rights
Source
Subject
Title
Type

"rightsHolder") or point to other aspects of the way an institutional context works. Operational fields might give information about how the object is related to other records or artifacts ("isPartOf" or "isReferencedBy"), or how it is to be used. Here, again, we see the difference between metadata that is descriptive and fields that are operational, defining relations among parts of a collection and/or its records.

In the humanities, many researchers have chosen to customize their metadata to serve particular purposes or to make a point about the materials under examination. The benefits and liabilities of customized metadata have to be weighed against the advantages of standards. Customized metadata structures are also common, particularly in digital humanities projects where the individual researcher or team has a specialized agenda for their work. This same work of art might be part of a study of gendered practices, concepts of artistic identity, or techniques of production. [See Exercise #2: Customized metadata.]

In digital work, the discussion of a "content model" or a "metadata scheme" allows a researcher to consider what features of a work or collection they consider important. If you are putting together materials that are related to each other by a particular theme—an event, site, or topic—thinking through the way they will be described is a significant feature of the intellectual work. In the case of an archive of queer performance, what fields in a metadata scheme allow that identity to be expressed? What types of performance need to be identified and in what vocabulary? Who defines that vocabulary? What are the politics of characterization of queer identity from within the community and from outside? Who speaks in the metadata? Categories that address ethnic and racial identity, sexual orientation, and gender, are particularly fraught since the act of classification can be oppressive.

Keeping metadata consistent is another challenge. Some of this is merely mechanical, such as making sure spelling and vocabulary remain the same so that if a record is being described as containing "colonial" terminology it does not vary into "colonialist" or "imperialist." One way to manage consistency is with "pick lists" that contain all of the vocabulary to be used in the fields of the metadata.

Creating highly customized and detailed metadata can be tempting. Specific expertise and research agendas lend themselves to nuanced and careful descriptive work. But the downside of such specialized activity is that the information may not work well if absorbed into larger systems. (See the discussion of Linked Open Data in Chapter 11 for related issues.) Idiosyncratically named and defined fields may not have a "target" if they are integrated into a system whose fields do not match. This is an old problem and was true for the upload of MARC into other systems as well. The migration of metadata is an ongoing problem.

Keep in mind that while metadata records are often considered a "surrogate" for a thing or object, a stand-in, they are also structured data and can be used for analysis. A study of catalog records, of acquisition records,

or any other kind of record can produce a portrait of a field across time. Metadata also provides a profile of institutional practices by tracking the changing descriptions of a collection. At what point, for instance, did the term *artist's book* start to be used to describe a particular kind of publication? Or *'zine*? The fields that are in medical records and census records have changed over time, as have diagnostic vocabularies for mental illness and pathology. *Where* a concept, artifact, or practice appears in a classification scheme can be as important as the terminology. Homosexuality used to be classified as a "pathology" and so books and materials relevant to it were described and positioned accordingly. Where are books about cooking placed (Hoffman 2013)? And what aspects of ethnic or national cuisine are identified? Classification schemes have their own history and though many standards persist, many have been questioned and reworked over time as values change.

Classification

One of the most powerful forms of organizing knowledge is through the use of classification systems—this is true for objects as well as documents. In digital environments (as in analog ones), classification systems are used in several ways—to organize the materials on a site, to organize files within a system, or to identify and name digital objects and/or the analog materials to which they refer. Classification systems impose a secondary order of organization into any field of objects such as texts, physical objects, files, images, recordings, etc. We use classification systems to identify and sort, but also to create models of knowledge. Knowledge models expose and embody cultural differences and values. These are implied in every act of naming or organizing. No classification system is value neutral, objective, or self-evident. All classification systems bear within them the ideological imprint of their production. A system of identifying works of art by their creators might be inappropriate in a community where practices are tied to tradition and repetition, rather than originality and invention.

Classification systems are closely related to metadata. The Dewey Decimal System and the Library of Congress are two different classification systems for published materials. They give each publication a unique identifier within their system and are used for shelving, storage, retrieval, and to position a book within the larger field of knowledge representation. The contrast between them is worth examining. The Library of Congress (LoC) has twenty-six divisions and uses the Roman alphabet as its structuring framework. It uses two of these, E and F, for the history of the Americas and only one, D, for the history of the rest of the world. While naval science and military science each have a dedicated category, caring professions do not appear, nor do activities traditionally associated with women's work, which are all subsumed under larger categories. Like LoC, the Dewey Decimal System begins with philosophy and religion, ends with geography and history

as one subsection, and reserves a special place for biography. Looking at a system like Dewey or LoC, one can discern a particular point of view about the world, the knowledge of it, and the values that shape understanding. Many of these systems were shaped by the influence of classical culture on western Renaissance thinkers (like the English scholar Francis Bacon) whose ideas are still legible in these schemes.

Classification systems arise in many fields. Carolus Linnaeus, the 18th century Swedish botanist, created a system for classifying plants according to their reproductive organs. Many of the relationships he identified and named have been contradicted by evidence of the genetic relations among species. Though his system is still used where its principles provide a uniform system, it has been supplemented by approaches that go beyond morphology (physical form). It can be surprising to find out who one's cousins are, genetically speaking. Next time you look in the mirror, see if you see a resemblance with a chicken, fruit fly—or a banana (Breakthrough 2017)! Is admitting to this a problem?

At the most basic level, we need classification systems to name and organize digital files. In addition, we use elaborate systems of naming and classifying that encode information about objects and/or knowledge domains.

Standards

Standardization is essential in classification systems. If you identify something as a "potato" one day and a "spud," or an "Idaho," or "Yukon gold" the next, how is someone to pick the ingredients for a recipe or manage inventory? This is especially true for a computer, which may not know that a "spud" can be turned into French fries. And if you list all your music recordings by artist's name and then list one by title, how will you find the anomalous item? Consistency is everything. When we are dealing with large scale systems used by many institutional repositories to identify and/or describe their objects, such as those mentioned above, then the necessity for standardization increases. If institutional repositories are going to be able to share information, that information has to be structured in a consistent and standardized manner, and it has to make use of standard vocabularies.

Standardization correlates to how information is used. Objects can be organized in an almost infinite number of ways. A collection of music recordings might be ordered by the length of the individual soundtracks, but this would make finding works by a particular artist, composer, or conductor impossible to locate.

Classification systems are used to organize collections, identify characteristics of objects in a system, and to name or identify those objects in a consistent way. They have a significant and substantive overlap with taxonomies and ontologies. Taxonomies are, quite literally, naming systems. They are comprised of selected and controlled vocabulary for naming items or objects. Ontologies are models of knowledge that organize information

and concepts into a structured system. There is no need to try to pin these words—classification, taxonomy, or ontology—into hard and fast definitions that are clearly distinct. They are not always distinct, and often resemble each other and are interchangeable with each other. In a general way, taxonomies are lists of terms/names, classification systems describe attributes and relations of objects in a system, and ontologies model knowledge systems. Folksonomies are created by user communities, rather than professionals, and sometimes these groups have specialized expertise, as in fan cultures. Confused? Don't be. Think of taxonomies as sets of names (What should I call it?), and ontologies as models of knowledge (Where does it belong in the scheme of things?).

Person and place names are also subject to standardization. The control of these terms is often linked to Name Authority files managed by professionals. Checking the Library of Congress for standard spellings will make your metadata more useful for exchange. The transliteration of information from foreign languages is styled according to different conventions over time. Being able to link all of these records has a benefit if one is trying to find information on, for instance, the Russian author Fyodor Dostoevsky, which can appear as: Dostoevski, Dostoevskij, Dostoevskii, Dostoyevsky, Dostoyevski, and so on.

Fluid ontologies: the politics of information vs. ideology of information

Before leaving the topic of ontologies, some discussion of the concept of fluid ontologies and politics of information is in order. Local and/or Indigenous knowledge of particular geographies, for instance, such as those of the Sami people who live in the arctic regions of the Northern hemisphere, were embodied in their place names. In some cases, these indicated potential hazards to a geographical feature (e.g. risk of avalanche), but this knowledge was erased in the classification imposed by Europeans. When disasters befell them, the Europeans realized the liabilities. At another extreme, local knowledge of a densely populated urban area and vernacular terminology for a sewer, ditch, or bridge might be essential for rescue and survival, while the managing municipality might have an abstract reference system disconnected from this information. A bulletin warning the population to avoid "Outflow pipe #337 in section A.45" might be meaningless, while an alert about "the lower Tremlow road" would not. The connection between these systems is sometimes referenced through the concept of "fluid" ontologies—the idea that a single naming system may be insufficient without cross-references (Wallack and Srinivasan 2009).

What is at stake in the use of classification and description systems, as well as the naming conventions? Costs (financial, cultural, and human) of mismatches between official and observed approaches to description of catastrophic events can be significant. Ontologies have cultural agency—they

both reflect and produce cognitive models of the world. In some cases, including adaptive, flexible tags that reflect local knowledge joined with the official meta-ontologies managed by the State or official cultural heritage institutions could be of value. This raises a question about how folksonomies and taxonomies/ontologies can be merged together. [See Exercise #3: Multiple ontologies.]

This brings the concept of folksonomies into play. Digital systems for tagging—whether attached to images, books, movies, bands, or any other category of cultural material—have proliferated in the web environment. Though great variety might exist in such tags—from the range of judgments to the specific vocabularies—the added value of folksonomies has been recognized by information professionals as well as the broader public. The question of how such information can be incorporated into institutional records is still a matter of discussion, but the fact that readers—expert or amateur—often make observations about materials that, if entered into records, would make these more findable and usable, is clear. Museums and other cultural institutions have been particularly interested in making use of such knowledge, but so are the tourist industry and entertainment realms.

Takeaway

Metadata is the term applied to information that describes information, objects, content, or documents. Metadata is used to classify, describe, organize, and connect records to artifacts and documents and to each other. Standard bibliographic metadata on library records includes title, author, publisher, place of publication, date, and some description of the contents, the physical features, and other attributes of the object. Metadata standards for research data will have other requirements for authenticating information and for making it able to be exchanged and used. Familiarity with metadata in a particular domain is essential for humanities research, whether the project takes place in universities, libraries, museums, and archives, or is produced for individual projects.

Exercises

Exercise #1: standard metadata

Try applying the Dublin Core metadata standard to documents or objects in your research field. Make records for half a dozen of the documents or artifacts. What information that is relevant to your work cannot be fit into this scheme? Which elements of the scheme are confusing or difficult to use consistently?

Exercise #2: customized metadata

Start creating a taxonomy and/or classification system for an area in which you have a high level of knowledge—such as your favorite music, family photographs, memorabilia, or other personal collection. What terms, references, or resources would you want to cross-reference repeatedly? Which should be in a pick list, so you could use them consistently? Is Dublin Core sufficient?

Exercise #3: multiple ontologies

What area of your own experience—health, emotional conditions, knowledge of a neighborhood or travel route, etc.—lends itself to the concept of "fluid" ontologies? Why would different naming conventions be of use?

Recommended readings

Hoffman, Gretchen. 2013. "How are Cookbooks Classified in Libraries?" *NASO* 4 (1). www.researchgate.net/publication/272962458_How_are_Cookbooks_Classified_in_Libraries_An_Examination_of_LCSH_and_LCC.

Pomerantz, Jeffery. 2015. "Introduction." In *Metadata*. Cambridge, MA: MIT Press. www.jstor.org/stable/j.ctt1pv8904.

4b Markup: XML, TEI, KML, JSON, and other standards

Markup languages are sets of codes, embodied in tags, that are used to analyze the contents of textual documents. They are also used to contain metadata about works and artifacts. More recently other formats, particularly JSON (JavaScript Object Notation), have become popular because they are more suited to data description.

HTML was already discussed as a standard for web publishing. HTML identifies the structural parts of documents and allows them to be displayed appropriately and styled. HTML is just one among many markup languages. At the highest level, these are all governed by rules in SGML, Standard Generalized Markup Language, first accepted as a standard in 1986. SGML is an overarching set of rules according to which these markup languages can be defined.[1] Think of SGML as instructions for how to make rules for specifying how to define document types. If this sounds confusing, think about how you would tell someone *how* to specify the rules that distinguish recipes from business letters from census forms. Different elements go into each, but rules about format and contents govern all of these as documents. Rules about how to make rules—the notion may make you dizzy. Perhaps this is a rare moment when a sports analogy is in order: the rules for baseball, basketball, and football are all radically different. But all have rules about scoring, errors, or fouls, and what is and isn't allowed. Imagine making a set of rules to be sure all sets of rules were certain to include those categories (specifying the kinds of rules needed for something to be a "sport"). SGML specifies what rules are needed for something to be a markup language.

The rules of SGML state that if you want to set up rules, the rules themselves have to conform to certain standards (they have be made of tags, tags can carry attributes, and have to be nested). In other words, all markup languages conform to certain guidelines of well-formedness. Individual markup languages are domain specific and will contain relevant components. Like metadata, markup languages are often created within industries or fields as standards so that, for instance, all medical publishing or legal documents can be governed by the same rules. This allows for exchange of information and analysis of large corpora.

A number of markup languages are commonly used in the humanities. These include XML (Extensible Markup Language), used for identifying semantic content; TEI (the standard of the Text Encoding Initiative); and KML (Keyhole Markup Language), used for geographical data. Because XML is easily customized—and human readable—with fields and tags specific to a project, it is used extensively for metadata and thus lends itself to interpretative work. XML schemes are often customized for a project so that description of interpretative features of the research can be formalized.

The most ubiquitous and familiar form of markup is HTML, already discussed. But HTML's purpose was to standardize the *display* of the *structure* of digital publications. HTML did not contain *semantic* tagging. Remember, structural tagging was concerned with the format of documents, not their content. It was focused on elements like headings, paragraphs, and so on. The need for *semantic* markup quickly became evident. If a corpus of six hundred articles was to be styled, HTML could not distinguish an author's name from a title, an article abstract from the body of the text, and a subtitle from a publisher's information when they were identified as "H1" or "H2". Those categories are *semantic* and their tags identify the content, not just the formal identity of the text. These are very fundamental distinctions, and semantic markup goes much further in practice. In fact, as will be seen, it becomes very fine-grained and has many "flavors" that have been developed for different disciplines. The standard format in which semantic mark-up is produced is XML and understanding what it does and how it works is essential for digital publishing and research.

As has already been mentioned, markup languages come in many flavors. Geospatial information uses KML, many text-based projects use TEI, and so on. The use of these standards helps projects communicate with each other and share data. A good exercise is to study a tag set for a domain in your area of interest or expertise and/or make one of your own. For instance, the creation of a specialized tag set allows people working in a shared knowledge domain to create consistency across collections of documents created by different users (e.g. Golf Markup Language, Music Markup Language, Chemical Markup Language, etc.). But a markup language is also a naming system, a way to formalize the elements of a domain of knowledge or expressions (e.g. texts, scores, performances, documents, etc.). [See Exercise #1: XML.]

In spite of the growing power of natural language processing (referred to as NLP), structured data remains the most common way of creating standards, formal systems, and data analysis. Structured data is particularly crucial as collections of documents grow in scale, complexity, or are integrated from a variety of users or repositories. Standards in data formats make it possible for data in files to be searched and analyzed consistently. (If one day you mark up Romeo and Juliet using the <girl> and <boy> tags and the next day someone else uses <man> and <woman> for the same characters, that creates inconsistency. In reality, the implementation of standards is difficult, inconsistency is a fact of life, and data crosswalks (matching values in one set of terms with those in another) only go partway toward fixing this problem (Zhang and Gourley 2009).

The standards for tags in markup languages, and their definition, rules for use, and other guidelines are maintained by the W3C (whom we have already discussed). However, they do not maintain the specifications for individual mark-up languages. All markup languages and structured data are subject to the rules of well-formed-ness. This means the files must be made so that they conform to the rules of markup to display properly, or "parse" in the browser. A file that does not parse is like a play made in a sport to which it does not belong (a home run does not "parse" in football) or a structure that is not correct (a circle that does not close, a square with six sides) because it does not conform to the rules.

TEI: a special subset

One form of XML is termed TEI for Text Encoding Initiative (which refers to the group of scholars who were responsible for creating it as a standard for marking up texts). As in the case of other professional organizations who maintain standards, the TEI is a consortium of nonprofit groups who support and guide its activities. Their guidelines were first released in 1994, and are meant, like other standards, to make digital documents interoperable. That means that the use of standard tags in accord with set rules and protocols allows these materials to be integrated into search, analysis, and research. While the TEI encourages use of standards, they also know that customization is an aspect of humanities research, and their structure allows for and supports this. [See Exercise #2: TEI.]

A quick look at the TEI Guidelines gives a sense of how closely it conforms to the study of bibliographical objects—books and manuscripts in particular. The major divisions of the TEI are Front Matter, Back Matter, and Body. The guidelines are specific to literary and humanistic texts, so there are standard tags for verse that include metrical analysis and rhyme, for instance, or metrical analysis within stanzas. By contrast, tag sets for marking up dictionaries include "entry" and "entry types" with the ability to specific cross-references, abbreviations, and other attributes. The tag sets become very elaborate to indicate every genre and format of text and

the many components specific to each one. So, the basic elements in any TEI document are things like quotation, punctuation, unusual language, editorial changes, and so forth, while manuscript description guidelines and performance scripts contain terms specific to these document types.

The first sample, above, shows bibliographical information. The TEI tags define fields for describing an edition. This is standard metadata. In the example below, TEI is used as "markup" to distinguish those parts of the text that are <state> directions from identification of a <speaker> to <l> lines being spoken, alternate readings in variant editions <rdg>, and other elements. The level of detail in the examples is useful for scholarly editions and detailed critical study but goes beyond the basics of metadata used for cataloging objects, books, or records.

TEI Guidelines have been issued in multiple languages in addition to English, including Chinese, Japanese, and Korean, as well as French, German, Italian, and Spanish. This gives a sense of how pervasive they are and why they are of value. But also, they have become widespread in their use and this, in turn, forces the community to take them into consideration and account.

Why would you use TEI? The circumstances in which it makes sense are determined by a few factors. If your project is institutionally based, then adhering to TEI will make the research more useful to others. If you are working on a topic of research that requires cooperation, this is essential. Projects with multiple partners or team members will find conformance a benefit, though interpreting what a particular tag means or requires can still be challenging. If you are working in a specialized area of research, TEI guidelines will structure the way you interpret the documents. For marking up women writers' work, you might want a set of feminist tags and terms. For analyzing documents of colonial politics, you might want others.

Table 4.2 TEI samples of bibliographical information

```
<sourceDesc>
 <biblStruct>
        <monogr>
        <author>Bandol </author>
        <title type="main">Metamorphosis</title>
        <title type="subordinate">Warped Tales</title>
        <edition>Second</edition>
        <imprint>
          <pubPlace>Rome</pubPlace>
          <publisher>Gambolini</publisher>
          <date>1782</date>
   </imprint>
   <extent>"32"</extent>
        </monogr>
 </biblStruct>
</sourceDesc>
```


Table 4.3 TEI markup of dramatic text

```
<stage type="entrance">Enter
        <name>Fairy</name>
        <name>Princess</name>
        and
        <name>Julianna</name>
</stage>


<sp who="prin">
<speaker>Princess</speaker>
        <l>Princess
        <name>Julianna</name>
        and lady
        <name>Jones</name>
        alive
        <app>
        <lem>?</lem>
        <rdg wit="#Q1 #Q2 #Q3 #Q4">,</rdg>
        </app>
        </l>
        <l>Welcome to
        <name>Fairyland:</name>
        the news is most welcome,</l>
        <l>That you are alive.</l>
</sp>
```

TEI is useful for formalizing the interpretation and study of texts. It forces the analysis of a literary work into a formal and explicit framework—is this statement this kind of thing or that? It adds value to a text by virtue of the intellectual labor in the markup that then can be subject to analysis in ways that mere string-searching does not permit. If a tag like "abuser" can be attached to proper names in a large corpus of legal documents, then this category can be searched, analyzed, and addressed. [See Exercise #3: Orlando Project.]

Markup is labor intensive. It cannot be automated very effectively. For many scholars, the activity of doing markup is a form of close reading that allows interpretation to be inserted into the body of the text. This approach has its limitations. Because TEI is based on XML, it is structured in nested hierarchies that cannot overlap. If a literary passage deals with multiple themes and ideas, the markup must be nested. But themes overlap throughout complex literary works, and so do motives or sentiments—which might be assigned to multiple characters across a wide variety of places in the text. For markup purposes, this overlap is not allowed, it will not parse, and the markup will simply have to contain the tags within each chapter. The representation of the text is not identical to its aesthetic structure. [See Exercise #4: TEI and title page elements.]

Other standards are also commonly used in the humanities. Another is Encoded Archival Description, or EAD. It is different from TEI in its

attention to the physical features of archival objects and it has its own vocabulary for description. OAI, the Open Archives Initiative is used to "harvest" information, particularly metadata, for integration into specialized, discipline-specific, search engines. Markup is ubiquitous in digital publishing and scholarship.

Another approach: JSON: describing objects rather than inserting markup

Markup is exceptionally well suited to the analysis and interpretation of textual elements, and thus used in literary and humanities work, as well as critical editing in all fields. XML formats are also used to create metadata, and thus to structure and describe the contents of records. But collections of documents lend themselves to other kinds of structured data description in file formats designed specifically for information exchange of data. The most popular of these is JSON format.

Not all data formats are text, and many files created in JavaScript contain information that cannot be readily transferred between a server and a browser and that won't fit easily into XML. JSON was developed to address this problem. It is text-based but can describe many data formats. It is thus a very useful standard with which to be familiar in order to facilitate exchange. JSON is used for storage as well as exchange and has the advantage that it is relatively simple to learn and is highly customizable.

JSON's notation scheme was designed so that it could describe data objects effectively. That means that its syntax contains ways to identify data types so they are not confused with each other. Also, JSON's syntax is very similar to that of Python and other languages, making it easy for those familiar with programming to learn and use it. Metadata, classification, and markup schemes have certain features in common. They are all ways of disambiguating data types, formats, and structures so that they can be processed and either analyzed or displayed properly—or exchanged and integrated into new projects. The crucial principle is that code of any kind conforms to formal rules that disambiguate data, content, and information in order to make their statements computable.

JSON is a simple but powerful way to take existing data formats and files and extract their information for the purpose of exchange or integration. Where XML and TEI are used to markup parts of texts—either thematically (an XML schema on genre might insert the term <mystery> or <thriller> as part of its analysis) or formally (a TEI scheme might contain the term <roleName> to provide highly specific information about an individual)—JSON syntax identifies types of structured data. It builds on JavaScript, but is a notation system (hence its name) for describing or representing what is in a data file. XML and JSON serve different purposes, but both are

components of digital research projects. If you are working with already structured content that contains complex data structures, files, or a range of formats, JSON is a useful way to describe it for transmission. JSON has a large user base and is likely to be supported as a format for the foreseeable future. These are also important considerations. [See Exercise #5: Contrast JSON and XML formats.]

Takeaway

Structured data comes in many forms. The choice to use markup languages or other formal methods of either analyzing or describing the contents of documents or files will depend upon the nature of the project, tasks to be performed, and use to which the information will be put. XML and JSON are both highly flexible and allow for considerable customization while also being of great value in storing and transmitting information. Subsets of XML are often discipline specific and provide standards for individual fields of expertise. The advantage of formal languages is that they disambiguate among components of text or data. Paradoxically, this is also their disadvantage—since so many humanistic documents or artifacts are filled with ambiguity that human readers have no difficulty processing. Computers, however, cannot tolerate ambiguity. This fundamental difference is worth keeping in mind. Is there an ethical issue in taking something that is fundamentally "both/and"—a metaphor and a literal statement, a fact and a project, a dream and a reality—and having to choose between representing it as one or the other? Absolutely.

Exercises

Exercise #1: XML

XML schema exist in many fields and disciplines. Look at this list, pick one, and begin to see how it identifies elements of documents for this field. While you are looking at the list, compare it to the Library of Congress and to the Dewey Decimal System. Why is it so different? https://en.wikipedia.org/wiki/List_of_types_of_XML_schemas

Exercise #2: TEI

TEI is a highly specialized XML standard for marking up literary and textual documents. It is divided by genre and its tags must be used in particular orders, hierarchies, and are highly defined. Compare the elements for 7. Performance Scripts, and 10. Manuscripts.

https://tei-c.org/release/doc/tei-p5-doc/en/html/index.html

Exercise #3: Orlando Project

Look at the documentation for the XML scheme and think about the customization of its tags and features. www.artsrn.ualberta.ca/orlando/

Exercise #4: TEI title page elements

Look at the elements for TitlePage in TEI www.tei-c.org/release/doc/tei-p5-doc/en/html/DS.html#DSTITL and think about how using these terms limits the kinds of materials that can be marked up. Are there publications that would not follow this form? Elements that might need to appear for it to work? What about artists' books? Or zines?

Exercise #5: contrast JSON and XML formats

Examine the contrasts in the JSON and XML formats of the same information. The main difference is the syntax and the way that the values are specified. Here are two excellent examples www.w3schools.com/js/js_json_xml.asp and https://json.org/example.html

Note

1 For those with an interest in rule-based systems and the highest level of document type specification, understanding the syntax of SGML boils down to a way to "declare" the contents of and the ways they can be used.

Recommended readings

"JSON vs XML." 2020. *W3schools*. www.w3schools.com/js/js_json_xml.asp.

Schwartz, Michelle, and Constance Crompton. 2018. "Remaking History: Lesbian Feminist Historical Methods in the Digital Humanities." In *Bodies of Information: Intersectional Feminism and the Digital Humanities*, edited by Elizabeth Losh and Jacqueline Wernimont, 131–56. Minneapolis, MN: University of Minnesota Press. Muse.jhu.edu/book/63017.

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Breakthrough Staff. 2017. "How Genetically Related Are We to Bananas?" *Breakthroughs*. www.breakthroughs.com/foundations-science/how-genetically-related-are-we-bananas.

Wallack, Jessica, and Ramesh Srinivasan. 2009. "The Local and the Global: Reconciling Mismatched Ontologies in Development Information Systems." HICSS Hawaii International Conference on Systems Sciences. 1–10.

Zhang Allison B., and Don Gourley. 2009. "Creating Metadata: Metadata Crosswalk." In *Creating Digital Collections*. Oxford: Chandos Publishing. www.sciencedirect.com/topics/computer-science/crosswalk.

Resources

DUBLIN Core https://dublincore.org/.

JSON (vs. XML) www.w3schools.com/js/js_json_xml.asp.

KML https://developers.google.com/kml/documentation/kml_tut.

Library of Congress Classification www.loc.gov/catdir/cpso/lcco/.

MARC Records www.loc.gov/marc/bibliographic/.

Mark-Up Languages (Wikipedia) https://en.wikipedia.org/wiki/List_of_XML_markup_languages.

Metadata Standards, European Union www.dcc.ac.uk/guidance/standards/metadata/list.

SGML www.w3.org/TR/WD-html40-970708/intro/sgmltut.html.

www.w3.org/MarkUp/SGML/.

TEI https://tei-c.org/.

www.tei-c.org/release/doc/tei-p5-doc/en/html/index.html.

Weather data and metadata standards 19th and 21st century versions https://mrcc.illinois.edu/data_serv/cdmp/cdmp.jsp.

www.almanac.com/weather/history/CA/Los%20Angeles/2018-07-02.

XML https://tei-c.org/release/doc/tei-p5-doc/en/html/SG.html.